

UME513 : Dynamics and Vibration

L	T	P	Cr
3	1	2*	4.0

Course Objective: This course deals with the dynamics of various physical systems like flywheels, gyroscopes etc. concepts of single-DOF and MDOF system vibrating systems. Moreover, the aim of this course is to model and analyze vibration of a-systems with their applications in the real world.

Syllabus

Dynamics: D'Alembert's Principle, Equivalent dynamical systems, Dynamic force analysis in engines. Turning moment diagram of the engines, Flywheel design, Study of motion transmission by Gear Trains.

Gyroscopes: gyroscopic action in automobiles, gyroscopes and their role in stabilization in ships, and airplanes.

Vibrations:

Single-DOF Systems: Free, forced, undamped and damped systems. Logarithmic decrement, Equivalent viscous and Coulomb damping, support excitation and vibrations due to unbalanced mass, vibration absorbers.

Two Degree of Freedom Systems: Free and Forced vibrations with and without damping, Principal and normal modes, coupling of modes.

Multi Degree of Freedom Systems: Various methods of analysis of multi degree freedom systems, numerical methods, Application to torsional vibrations, Modal Analysis and its applications to realistic problems.

Introduction to vibration of continuous system: Vibration of Beam/Bar

Research Assignment: Group assignment for this course may include one of the following topics:

- Natural frequencies of physical systems
- Modal analysis of realistic systems
- Suspension systems of vehicles
- Vibration isolation of machines
- Gyroscopic effect in aero planes and ships

Micro Project: Group project for this course may include one of the following topics:

- Determine the natural frequencies of physical multi DOF systems like, Beam, suspension system, bridge etc.
- Calculate the mode shapes and perform modal analysis analytically and validate the results obtained using commercial packages.
- Determination of damping value experimentally using logarithmic decrement method and validate theoretically.
- Develop simple 1-2 DOF system like inverted pendulum, measure the system response and relate to theoretical concepts for validation.

Laboratory Work

Basic knowledge and experiments related to simple pendulum, compound pendulum, damping coefficient, critical speed, Data acquisition using accelerometers.

Course Learning Objectives (CLO)

The students will be able to:

1. Apply engineering principle of mechanics to design motion transmission devices and flywheels.
2. Determine the appropriate parameters for stability a system using the concept of gyroscopic action.
3. Derive the dynamic model of real-life problems and verify the natural frequencies and mode shapes.
4. Analyze two- and multi-DOF physical systems analytically and validate using a commercial package

Text Books

1. Rao, S. S., Mechanical Vibrations, Addison Wesley Publishing Company, New York (1995).
2. S. S. Rattan, Theory of Machines, Tata Mc Graw Hill (2012).
3. J. J. Uicker, G. R. Pennock, and J. E. Shigley, Theory of Machines and Mechanism, Oxford Press (2009)
4. Grover, G. K., Mechanical Vibrations, Nem Chand and Bros, Roorkee (1996).

Reference Books

1. K. Ghosh and A. K. Mallik, Theory of Mechanisms and Machines, Affiliated East-West press Pvt. Ltd. (1993)
2. Kelly, S. G., Mechanical Vibrations, Schaum's Outlines, Tata McGraw Hill, New Delhi (2007).
3. Srinivasan, P., Mechanical Vibration Analysis, Tata McGraw Hill, New Delhi (1995).

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	40
3.	Sessionals (Minor Project/Tutorials/Quizes)	30